
AMPERE DEVELOPMENT

Power Testing & Load Bank Solutions

Rent vs. Buy Load Banks

A Decision Worksheet

When Rental Beats Ownership for Load Testing — and When It Doesn't

This is a practical worksheet, not a sales essay. Use the break-even math, the decision matrix, and the hidden-cost list to decide whether ownership or rental is the better fit for your generators, UPS systems, data-center commissioning, or multi-site testing program. Ampere sells rental — and we will tell you when buying is smarter.

Honesty converts better than “always rent.” Fill in your numbers. Score the matrix. Then call us with a clear answer — or a clear question.

How to Use This Worksheet

1. Complete the **Break-Even Math** page with your estimated purchase price and annual ownership costs, then compare to expected annual rental spend.
2. Score the **Decision Matrix**. More “Favors rent” answers usually mean rental wins on total cost and risk.
3. Read the **Hidden Costs of Ownership** so nothing is left off the balance sheet.
4. Check Ampere's honest guidance on when we recommend buy vs. rent.
5. Use the final page checklist to request an accurate rental quote.

What This Worksheet Does Not Do

It does not replace a formal quote, tax advice, or a full capital-budget analysis. Prices vary by capacity, voltage, resistive vs. reactive vs. DC, medium-voltage, liquid-cooled, delivery distance, and market conditions. Treat the example numbers as starting points; replace them with your quotes.

1. Break-Even Math

Ownership cost is not just the purchase price. Rental cost is not just the weekly rate. Compare full annual cost of ownership against full annual cost of rental for the same testing program.

A. Ownership Cost (Annualized)

Use a planning horizon of 5–7 years for most resistive load banks (longer only if utilization is truly high and technology is stable).

Ownership Cost Element	Example (illustrative)	Your Number
Purchase price (load bank only)	\$60,000 – \$120,000 (500 kW–1 MW class)	_____
Cables, connectors, trailer / skid mods	\$5,000 – \$25,000+	_____
Delivery / commissioning of owned unit	\$1,500 – \$5,000	_____
Storage (space + climate / security) / yr	\$1,200 – \$6,000+	_____
Preventive maintenance + repairs / yr	\$1,500 – \$5,000+	_____
Calibration / verification / yr	\$500 – \$2,000	_____
Insurance / property tax on asset / yr	\$400 – \$2,000	_____
Internal labor to operate & move / yr	\$1,000 – \$8,000	_____
Transport between sites (if multi-site) / yr	\$0 – \$10,000+	_____
Capital cost of money (or opportunity cost) / yr	5–10% of purchase	_____

Simple annual ownership estimate: (Purchase + cables + setup) ÷ years of useful life + annual operating costs above.

Example: \$90,000 all-in purchase ÷ 6 years = \$15,000/yr capital recovery + \$6,000/yr operating = **~\$21,000/year** before major repairs or obsolescence.

B. Rental Cost (Annualized)

Rental Cost Element	Example (illustrative)	Your Number
Rental rate × days/weeks needed / yr	e.g. 2 weeks × \$2,000–\$5,000	_____
Delivery + pickup / mobilization	\$500 – \$3,000+ per trip	_____
Cables / accessories (if not included)	Often included or modest add-on	_____
On-site tech / setup support (optional)	\$0 – \$2,000+ per event	_____
Damage waiver / insurance (if elected)	Per contract	_____

Example: One annual 5-day test + delivery ≈ **\$3,000–\$8,000 total** depending on size and location. Two project mobilizations per year might still total well under the annual ownership estimate above.

C. Break-Even Check

Rough rule: if you need the equipment less than roughly **8–15 weeks per year** at the same site and same configuration, rental usually wins on pure cost — and almost always wins on flexibility and risk. High utilization (weekly/monthly at one fixed voltage) can favor ownership.

Your break-even weeks = (Annual ownership cost) ÷ (Weekly rental all-in rate). Fill in your numbers before deciding.

My estimated annual ownership cost: \$ _____ | My estimated annual rental cost: \$ _____

Decision from the math: ■ Rent looks better ■ Buy looks better ■ Too close — use the matrix

2. Decision Matrix

Circle or mark the column that matches your situation for each factor. Count the marks. A clear majority on one side is a strong signal. Mixed scores mean dig deeper on the math and hidden costs.

Factor	Favors Rent	Favors Buy	Your Side
Frequency of use	Annual / project-based / seasonal	Weekly-monthly, same site	■ R ■ B
Voltage / kW mix	Changes by job	Always the same	■ R ■ B
Locations	Multi-site	One plant / fixed location	■ R ■ B
Duration of each need	Days to a few weeks	Months, dedicated unit	■ R ■ B
Staff / operators	No dedicated techs or operators	In-house capability	■ R ■ B
Technology change	AI racks, 800 V DC, new densities, MV	Stable, unchanging requirement	■ R ■ B
Capital priority	Preserve cash / prefer OpEx	Capital available, want asset	■ R ■ B
Peak capacity need	Occasional multi-MW or specialty	Steady mid-size only	■ R ■ B
Obsolescence risk	High (standards & power density move)	Low (fixed test profile)	■ R ■ B
Utilization certainty	Uncertain future testing volume	Predictable high utilization	■ R ■ B

Rent score: _____ | Buy score: _____ | Lean: ■ Rent ■ Buy ■ Hybrid / more analysis needed

How to Read the Matrix

- **6+ “Favors rent”** → Rental is usually the rational choice on cost, flexibility, and risk.
- **6+ “Favors buy”** and high utilization at one fixed configuration → Ownership can make sense; still run the math.
- **Split score** → Look at break-even weeks and hidden ownership costs before committing capital.
- Technology-change and multi-site factors are often decisive even when pure cost is close.

3. Hidden Costs of Ownership

Purchase price is the visible number. These items are the ones that quietly erase the “savings” of ownership when utilization is moderate or low.

Storage & Real Estate

- Indoor or protected outdoor space, security, and access for a large, heavy asset that sits idle most of the year.
- Climate and condensation control for electronics and fan motors (idle outdoor storage is a known failure mode).
- Opportunity cost of the square footage that could generate revenue or reduce other storage spend.

Cables, Connectors & Accessories

- Correct ampacity, length, and connector sets for every voltage you test — often a second major purchase.
- Wear, theft, and damage; replacement and inventory management over years.
- Specialty gear for medium-voltage, DC, or unusual interfaces that you only need occasionally.

Maintenance, Calibration & Downtime

- Annual or periodic calibration so readings remain trustworthy for compliance and acceptance.
- Fan motors, contactors, resistors, controls — failures happen; you own the repair lead time.
- When **your** load bank is in the shop, your test schedule slips unless you rent anyway.

Insurance, Compliance & Obsolescence

- Property insurance and any applicable tax on the asset.
- Technology moves: AI rack densities, 800 V DC, liquid-cooled testing, new IST methods. An owned bank can lag.
- You still may need to rent specialty units for wet-stacking cleanup at higher loads, reactive power factor, MV, or multi-MW peaks — so ownership does not eliminate rental.

People & Logistics

- Trained operators, safety procedures, and time spent moving and connecting equipment.
- Multi-site transport, permits, and coordination — costs that scale with geography, not with ownership pride.

Reality check: Many organizations that “saved money by buying” still rent several times a year for capacity, voltage, or specialty needs they did not anticipate — and still carry storage and maintenance on the owned unit.

4. When Ampere Says “Buy”

We sell rental. We still tell customers to buy when the facts support it. Ownership tends to win when most of the following are true:

- **Same voltage and capacity** every time — no mix of 480 V, 208 V, MV, DC, or reactive needs.
- **One plant or fixed site** — the unit rarely or never leaves.
- **High utilization** — weekly or monthly testing, or continuous process use, so capital recovers quickly.
- **In-house technical staff** who can operate, maintain, and troubleshoot the bank.
- **Stable technology requirement** for the next 5+ years — no major shift in power density, voltage class, or test method expected.
- **Capital is available** and you prefer an on-balance-sheet asset with depreciation benefits that fit your tax plan.

If those conditions hold, ask us for a rental quote anyway for the first year of projects while you evaluate suppliers for purchase — or consider a rent-to-own path if available.

5. When Ampere Says “Rent”

Rental is usually the better answer when any of the following dominate:

- **Commissioning & IST / Level 5** — one-time or infrequent full-facility tests.
- **Seasonal or annual compliance** — NFPA 110-style tests, hospital cycles, storm-season readiness.
- **Multi-site programs** — different plants, campuses, or project locations.
- **Medium-voltage** testing — specialized, expensive to own for occasional use.
- **Liquid-cooled or high-density data-center** simulation — equipment that changes with each generation of racks.
- **800 V DC, electronic / programmable, or specialty** loads — high capital, fast technology turnover.
- **Variable kW / voltage mix** — each job needs a different bank size or type.
- **No dedicated operators or storage** — you want the asset and the logistics off your plate.
- **Uncertain future volume** — you do not want to bet capital on testing frequency that may drop.

In short: **buy when the need is fixed, frequent, and local. Rent when the need is variable, project-driven, multi-site, or specialized.**

6. What to Send Us for a Quote

Accurate quotes need accurate inputs. Copy or complete this checklist and send it to sales@ampere.dev or call 888-345-LOAD.

Project / Test Basics

- Purpose: commissioning / IST / annual compliance / wet-stacking cleanup / UPS-battery / other: _____
- Preferred test dates or window: _____ Duration on site (days): _____
- Site address / city / state: _____
- Access constraints (hours, permits, crane, trailer limits): _____

Electrical Requirements

- Required kW / kVA: _____ Voltage(s): _____ Frequency: 50 / 60 Hz
- Load type: resistive / resistive-reactive / capacitive / DC / electronic-programmable / other
- Power factor target (if reactive): _____ Medium-voltage? Yes / No — kV class: _____
- Connection interface (cam-lok, bus bar, other) and cable length needed: _____

Unit Under Test

- Generator / UPS / battery / data-center hall / solar-BESS / transformer / other: _____
- Nameplate ratings and quantity of units: _____
- Any paralleling, transfer-switch, or building-load coordination notes: _____

Support & Logistics

- Delivery only / delivery + setup assistance / full tech support during test
- On-site contacts and phone numbers: _____
- Special safety or site rules we should know: _____

Your Decision Summary (optional — for your records)

Break-even math leans: ■ Rent ■ Buy | Matrix leans: ■ Rent ■ Buy | Final call: ■ Rent ■ Buy ■ Hybrid

Notes: _____

Ready for numbers that match your site?

Send the checklist above — or just call. We will quote the rental that fits,
and we will tell you if ownership looks smarter for your case.

888-345-LOAD · ampere.dev · sales@ampere.dev

Illustrative cost ranges in this worksheet are planning aids only. Actual purchase and rental prices vary by manufacturer, capacity, configuration, market, and logistics. Always confirm with current quotes.